

Vivek Kumar

SOFTWARE ARCHITECT / PRINCIPAL ENGINEER

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PROFESSIONAL SUMMARY

Software architect and engineering leader with **14+ years of** building large-scale systems across airline, media, and publishing domains. I break big monolithic applications into smaller, independent services, shape how systems talk to each other, and build-in security and scale from the start. Known for setting technical direction, mentoring teams, and turning complex business needs into simple, maintainable solutions.

CORE COMPETENCIES

Architecture & Design	Microservices · Event-Driven Architecture · Distributed Systems · Domain-Driven Design · API Gateway · Sidecar Pattern · System Design & Scalability
Backend	Java 17 · Spring Boot · Spring Cloud · Hibernate · Struts 2 · REST APIs
Frontend	Angular · React.js · Vue.js · Micro-frontends · Module Federation · Webpack
Messaging & Data	Apache Kafka · Solace PubSub+ · Couchbase · Oracle · MySQL · Apache Solr
Platform & Practice	Token-based Auth · CI/CD · Jenkins · Git · VAPT / PT Remediation · Agile / Scrum · Mentoring

PROFESSIONAL EXPERIENCE

Coforge

Senior Technical Lead · Software Architect track | Promoted from Technology Specialist · Aug 2022 – Present
· Client: Emirates, Greater Noida

- Lead the architecture and technical direction for large airline platforms (Emirates), guiding system design and a team of 7+ developers and QA engineers.
- Led the move of a large monolithic booking platform (ResConnect) into smaller, independent services, splitting the frontend and backend so each can be built and released on its own.
- Designed domain-driven, stateful microservices for the Booking and Ancillary domains using Java 17 and Spring Boot.
- Designed the API gateway and security setup, with one central place to check access tokens so services can talk to each other safely.
- Used event-based messaging (Apache Kafka and Solace) to signal when screen recording should pause and resume — pausing while a customer, or an agent acting on their behalf, enters payment details, so sensitive information is never captured. Used Couchbase for fast, distributed data access.
- Strengthened platform security by fixing vulnerability and penetration-test findings and applying secure practices across services.
- Built a proof-of-concept for modernizing the frontend into modular pieces, and mentoring engineers through design reviews, code reviews, and key decisions.

Hindustan Times

Senior Software Engineer | Sep 2010 – Aug 2022

- Designed and delivered large content publishing platforms, owning the database design, backend services, and search.

- Built scalable REST APIs and backend systems with Spring, Struts 2, and Hibernate; designed Apache Solr search with indexing and query optimization for high-volume content.
- Engineered automated content-aggregation crawlers (Selenium) across 200+ sources and data pipelines for ingestion, transformation, and delivery of articles, images, and media.
- Developed CMS web interfaces with Vue.js and AngularJS. Optimized MySQL queries and indexing for high-volume processing.
- Led a small team, ensuring code quality, system stability, and timely delivery.

Thomson Digital — Thomson Press (India) Ltd.

Executive – R&D | Jun 2007 – Jul 2010

- Designed XML-based automation workflows for publishing systems, automating Adobe InDesign typesetting via the JavaScript SDK.
- Built XML transformation pipelines using Java, XSLT, and XPath, plus QA automation for validation and content-quality checks.

KEY ACHIEVEMENTS

- **Zero-impact migration** of stateless services from TIBCO to Solace using microservices + feature-flag strategy with no system downtime — recognized by the client.
- Led a step-by-step shift from one large application to smaller services, shaping the service design that the whole platform was built on.
- Adopted AI tooling for unit-test generation and legacy-code comprehension, measurably improving delivery efficiency.

EDUCATION

B.Sc. in Computer Science — Sikkim Manipal University

Diploma in Information Technology — Board of Technical Education, Uttar Pradesh

PROJECT & FEATURE DETAILS

Airline Agent Booking Platform — ResConnect

Senior Technical Lead / Architect | Coforge · Emirates

Modernization of the airline agent booking system — turning a large legacy application into a set of smaller, independent services that are easier to scale, secure, and release.

FEATURES & CONTRIBUTIONS

- **Booking & Ancillary services:** designed independent services for the core booking and add-on (ancillary) domains using Java 17 and Spring Boot.
- **API gateway & security:** built a single entry point for routing requests, with one central place to check access tokens so services can talk to each other safely.
- **Payment-safe screen recording:** used event-based messaging (Kafka / Solace) to automatically pause recording while a customer, or an agent acting on their behalf, enters payment details — then resume afterwards, so card data is never captured.
- **Frontend & backend split:** separated the frontend and backend into independent codebases so each team can build and release on its own.
- **Micro-frontend POC:** built a proof-of-concept (Module Federation) to break the UI into modular pieces that can evolve independently.
- **Security hardening:** resolved vulnerability and penetration-test findings and applied secure-by-design practices across services.
- **Zero-downtime migration:** moved stateless services from TIBCO to Solace using a feature-flag approach with no impact to the running system — recognized by the client.

Stack: Java 17 · Spring Boot · Spring Cloud Gateway · Kafka · Solace · Couchbase · React.js · AngularJS · CI/CD

Content Ingestion & Publishing Platform

Senior Software Engineer | Hindustan Times

Large-scale platform to gather, process, and publish news content — from automated collection through search and delivery.

FEATURES & CONTRIBUTIONS

- **Search system:** built fast content search with Apache Solr, including indexing and query tuning for high volumes of articles.
- **Automated content collection:** developed crawlers (Selenium) that gather content automatically from 200+ sources.
- **Data pipelines:** created pipelines to ingest, transform, and deliver articles, images, and media; managed FTP-based distribution.
- **CMS interfaces:** built web interfaces with Vue.js and AngularJS for managing content.
- **Performance:** optimized MySQL queries and indexing for high-volume processing.

Stack: Java · Spring · Struts 2 · Hibernate · MySQL · Apache Solr · Selenium · Vue.js · AngularJS

Publishing Workflow Automation

Executive — R&D | Thomson Digital

Automation of publishing workflows using XML-based systems and Adobe InDesign, reducing manual typesetting effort.

FEATURES & CONTRIBUTIONS

- **InDesign automation:** automated page typesetting using the Adobe InDesign JavaScript SDK.
- **Transformation pipelines:** built XML transformation pipelines using Java, XSLT, and XPath.
- **Quality automation:** created scripts for XML validation and content-quality checks.

Stack: Java · XML · XSLT · XPath · Adobe InDesign SDK